New proper circulant weighing matrices from a twin of the Singer construction

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Abstract

A circulant weighing matrix $\mathrm{CW}(n,k)$ is a circulant $n\times n$ matrix W with entries in $\{0,\pm1\}$ and $WW^{\mathsf{T}}=kI$; it is proper if it is not the lift of a smaller one. We construct proper $\mathrm{CW}(146,8^2)$, $\mathrm{CW}(266,11^2)$, $\mathrm{CW}(366,13^2)$, $\mathrm{CW}(504,12^2)$, $\mathrm{CW}(614,17^2)$ and $\mathrm{CW}(762,19^2)$. To our knowledge, as of October 8, 2026, the existence of a proper circulant weighing matrix with each of these parameters was open: all six are listed as open in Gordon's La Jolla Circulant Weighing Matrix Repository and none appears in Gordon's 2026 table of known proper $\mathrm{CW}(n,k)$ with $n<1000$, $k\le 19^2$. For $\mathrm{CW}(504,144)$ no circulant weighing matrix of any kind was known. All six first rows are explicit and are verified in Lean 4 (core library only) by kernel evaluation. Four of them come from a new "twin" of the classical Singer-complement construction. For q odd, the standard $\mathrm{CW}(q^2+q+1,q^2)$ on the points of $\mathrm{PG}(2,q)$ has a twin, obtained by composing with the homology of ratio -2 (an elation when $p=3$), that has the same support. Whenever the twin is again a circulant weighing matrix, a doubling lemma produces a proper $\mathrm{CW}(2(q^2+q+1),q^2)$. We conjecture that this holds for every odd prime power q. It has been checked by computer for all odd prime powers $q\le 251$, but we have no proof. The other two matrices come from a multiplier-restricted search ($n=146$) and from a product over non-coprime orders ($n=504$).

Contents

1 Introduction **2**

2 Preliminaries **3**

3 Three elementary lemmas **4**

4 The twin of the Singer-complement construction **4**

5 A proper $\mathrm{CW}(146,8^2)$ **7**

6 A proper $\mathrm{CW}(504,12^2)$ **7**

7 Formal verification **7**

8 What is not verified; open questions **9**

A First rows of the six new matrices **9**

1 Introduction

A *circulant weighing matrix* $\mathrm{CW}(n,k)$ is an $n \times n$ circulant matrix with entries in $\{0,1,-1\}$ such that $WW^{\mathsf{T}} = kI_n$. It is determined by its first row $a = (a_0, \ldots, a_{n-1})$, and the condition says exactly that a is a *perfect ternary sequence*: its periodic autocorrelation

$$\mathrm{PAF}_a(d) = \sum_{i=0}^{n-1} a_i \, a_{(i+d) \bmod n}$$

equals k for $d = 0$ and 0 for $0 < d < n$. Necessarily $k = s^2$ is a square. If W is a $\mathrm{CW}(m,k)$ and $m \mid n$, then spreading its first row out (the group-ring map $X \mapsto X^{n/m}$) gives a $\mathrm{CW}(n,k)$. Matrices that do not arise in this way are called *proper*: the support of the first row is not contained in a coset of a proper subgroup of $\mathbb{Z}_n$.

Strassler's 1997 table of existence results for $n \le 200$, $k \le 100$ was updated by Tan and reduced to 27 open cases by Arasu, Gordon and Zhang [1]. Tan [10] recently settled eight more cases (all nonexistence results). Gordon maintains the La Jolla Circulant Weighing Matrix Repository [4]. It records the existence status of proper $\mathrm{CW}(n, s^2)$ for all $n \le 1000$, $s \le 19$, together with explicit matrices. Table 7 of [3], reproduced on the repository web page (last updated 2026-09-03), lists all *known* proper $\mathrm{CW}(n,k)$ with $n < 1000$ and $k \le 19^2$. The web page remarks that for larger k almost all entries come from the product or relative-difference-set (RDS) constructions, "so presumably many others remain to be found". For odd prime-power weight k, Leung and Schmidt [7] proved that only finitely many proper $\mathrm{CW}(n,k)$ exist (for $k = 4$ there are infinitely many, so the parity condition is needed). Leung and Ma [8] asked (open problem 7 in the slides of a 2011 talk): apart from the classical construction of proper $\mathrm{CW}((p^2+p+1)w/2, p^2)$ ($w \mid p-1$, $w \equiv 2 \pmod 4$), are there other (infinite) families of proper $\mathrm{CW}(v, p^2)$?

A further piece of history is relevant here. Theorem 6.6 of [1] asserted the existence of a proper $\mathrm{CW}((q^3-1)/n, q^2)$ for every divisor $n > 1$ of $q-1$ when q is odd. Its proof used an RDS-to-CW theorem (Theorem 6.4 there) which, as Gordon later pointed out [3, Theorem 12], needs both indices m and n of the $(m, 2n, k, \lambda)$-RDS to be odd. Gordon's corrected version (Theorem 13 of [3]) only yields orders $(q^2+q+1)n'$ with n' odd. Accordingly, the 2025/26 table no longer lists 366 (13^2), 614 (17^2) or 762 (19^2), all of which were listed in Table 11 of [1]. The case $n = (q-1)/2$ of the withdrawn claim is precisely the family $\mathrm{CW}(2(q^2+q+1), q^2)$ studied here.

Results. Table 1 lists the six parameter pairs settled here.

Theorem 1.1 (main theorem; Lean-certified)**.** *There exist proper circulant weighing matrices* $\mathrm{CW}(146, 64)$*,* $\mathrm{CW}(266, 121)$*,* $\mathrm{CW}(366, 169)$*,* $\mathrm{CW}(504, 144)$*,* $\mathrm{CW}(614, 289)$ *and* $\mathrm{CW}(762, 361)$*.*

Conjecture 1.2 (twin conjecture; Conjecture 4.6 below)**.** *For every odd prime power q the twin $W_{1/2}$ of the classical* $\mathrm{CW}(q^2+q+1, q^2)$ *is again a circulant weighing matrix. Consequently a proper* $\mathrm{CW}(2(q^2+q+1), q^2)$ *exists for every odd prime power q.*

The conjecture has been verified for every odd prime power $q \le 251$ (62 values; see Remark 4.8). For $q = 3, 5, 7, 9$ it recovers the orders $26, 62, 114, 182$, which [3] lists as sporadic (boxed) entries. If true, it would answer the question of Leung and Ma with a new infinite family. We stress that we do *not* have a proof.

(n,k)	repository [4]	Table 7 of [3]	non-proper CW known?	construction
$(146,8^2)$	Open	no	yes (lift of $\mathrm{CW}(73,8^2)$)	multiplier search, §5
$(266,11^2)$	Open	no	yes (lift of $\mathrm{CW}(133,11^2)$)	twin, $q=11$, §4
$(366,13^2)$	Open	no (dropped)	yes (lift of $\mathrm{CW}(183,13^2)$)	twin, $q=13$
$(504,12^2)$	Open	no	no	product, §6
$(614,17^2)$	Open	no (dropped)	yes (lift of $\mathrm{CW}(307,17^2)$)	twin, $q=17$
$(762,19^2)$	Open	no (dropped)	yes (lift of $\mathrm{CW}(381,19^2)$)	twin, $q=19$

Table 1: The six parameter pairs. "Open" is the status in the repository data file (v1.3, commit 5b3d729, 2026-04-24); the second parameter there is $s=\sqrt{k}$. "Dropped" means the entry appeared in [1, Table 11] but not in [3, Table 7].

Novelty and caveats. We compared with [4, 3, 1, 10, 7, 9] and with the perfect-ternary-sequence literature that we could access (Ipatov's construction as described in [5], and [5] itself). We could *not* check Krengel [6], which (according to its abstract) constructs almost-perfect ternary sequences of length $(p^n-1)/r$ and odd-perfect ternary sequences of half that length from m-sequences, in particular odd-perfect sequences of length $2(p^n-1)/(p^{n/3}-1)=2(q^2+q+1)$. Odd-perfect sequences are not perfect, so these are different objects from our $\mathrm{CW}(2(q^2+q+1),q^2)$; but a perfect sequence of odd length m becomes odd-perfect after the sign change $i\mapsto(-1)^i$, so the twin $W_{1/2}$ itself might appear there in another guise. Our novelty claims are therefore "to our knowledge, as of October 8, 2026".

2 Preliminaries

We identify $\mathbb{Z}_n$ with the cyclic group $\langle X\rangle$, $X^n=1$, and a sequence a with $A=\sum_i a_iX^i\in\mathbb{Z}[\mathbb{Z}_n]$. For $A=\sum_g a_g g$ in a group ring $\mathbb{Z}[G]$ of a finite abelian group G we write $A^{(-1)}=\sum_g a_g g^{-1}$ and $\operatorname{supp}A=\{g:a_g\neq 0\}$. Then a is the first row of a $\mathrm{CW}(n,k)$ if and only if $a_i\in\{0,\pm1\}$ and $AA^{(-1)}=k$. We say A is a CW in G if this holds with G in place of $\mathbb{Z}_n$.

Definition 2.1. $A\in\mathbb{Z}[G]$ is *proper* if $\operatorname{supp}A$ is not contained in a coset $g+L$ of a proper subgroup $L<G$. For $G=\mathbb{Z}_n$ the subgroups are $e\mathbb{Z}_n$ ($e\mid n$), so A is proper if and only if for every divisor $e>1$ of n the support meets at least two residue classes modulo e.

Lemma 2.2 (Fourier criterion)**.** *For $A\in\mathbb{Z}[G]$ we have $AA^{(-1)}=k$ if and only if $|\chi(A)|^2=k$ for every character χ of G.*

This is standard: $\chi(AA^{(-1)})=|\chi(A)|^2$, and an element of $\mathbb{C}[G]$ is determined by its character values.

Field notation. Let q be an odd prime power, $F=\mathbb{F}_q$, $K=\mathbb{F}_{q^3}$, and let $\mathrm{T},\mathrm{N}\colon K\to F$ be the trace and norm. Let η be the quadratic character of F (with $\eta(0)=0$) and $\eta_K=\eta\circ\mathrm{N}$; this is the quadratic character of K, because $\mathrm{N}\colon K^*\to F^*$ is onto. Put $m=q^2+q+1$ and $G=K^*/F^*$, a cyclic group of order m. Its elements are the points $[y]$ of $\mathrm{PG}(2,q)$, where K is viewed as F^3. Fix a primitive element α of K and identify $i\in\mathbb{Z}_m$ with $[\alpha^i]$; under this identification multiplication in K^* becomes the Singer cycle. Let $L=\{[y]:\mathrm{T}(y)=0\}$, a line of $q+1$ points (the classical Singer difference set), and let $S=G\setminus L$, so $|S|=q^2$. Every point of S has exactly one representative in the affine plane $P=\{y\in K:\mathrm{T}(y)=1\}$.

3 Three elementary lemmas

Lemma 3.1 (doubling)**.** *Let m be odd. Identify $\mathbb{Z}_{2m} \cong \mathbb{Z}_m \times \mathbb{Z}_2$ by $x \mapsto (x \bmod m,\ x \bmod 2)$, and let z be the element of order 2. For $U, V \in \mathbb{Z}[\mathbb{Z}_m]$ put $A = U + zV$. Then $AA^{(-1)} = k$ if and only if $(U+V)(U+V)^{(-1)} = k$ and $(U-V)(U-V)^{(-1)} = k$.*

Proof. Every character of $\mathbb{Z}_m \times \mathbb{Z}_2$ is $\chi \otimes \epsilon$ with $\epsilon(z) = \pm 1$, and $(\chi \otimes \epsilon)(A) = \chi(U \pm V)$. Apply Lemma 2.2 in $\mathbb{Z}_{2m}$ and in $\mathbb{Z}_m$. □

Corollary 3.2. *Let m be odd and let W, W' be first rows of two $\mathrm{CW}(m,k)$ with the same support. Define $A \in \{0, \pm 1\}^{2m}$ by $A_x = \frac{1}{2}(W + W')_{x \bmod m}$ for even x and $A_x = \frac{1}{2}(W - W')_{x \bmod m}$ for odd x. Then A is a $\mathrm{CW}(2m,k)$. If in addition W is proper and $W' \neq \pm W$, then A is proper.*

Proof. Since $W'_i = \pm W_i$ on the common support, $U = \frac{1}{2}(W + W')$ and $V = \frac{1}{2}(W - W')$ have entries in $\{0, \pm 1\}$ and disjoint supports. Under the identification of Lemma 3.1, $A = U + zV$, and $U \pm V \in \{W, W'\}$, so A is a $\mathrm{CW}(2m,k)$.

For properness, let $e \mid 2m$ with $e > 1$ and suppose $\operatorname{supp} A \subseteq c + e\mathbb{Z}_{2m}$. If e has an odd prime factor p, then $p \mid m$ and every $x \in \operatorname{supp} A$ satisfies $x \equiv c \pmod p$. Hence $x \bmod m$ lies in one residue class mod p, so $\operatorname{supp} W$ lies in a coset of $p\mathbb{Z}_m$, contradicting properness of W. Otherwise $e = 2$ (as $4 \nmid 2m$), so $\operatorname{supp} A$ lies in one parity class, i.e. $U = 0$ or $V = 0$, i.e. $W' = \pm W$. □

Lemma 3.3 (product over non-coprime orders)**.** *Let H, K' be subgroups of a finite abelian group G, and let $A \in \mathbb{Z}[H]$, $B \in \mathbb{Z}[K']$ be* CW*s in H and K' of weights k_1, k_2. Assume*

$$(\operatorname{supp} A - \operatorname{supp} A) \cap (\operatorname{supp} B - \operatorname{supp} B) = \{0\}.$$

Then AB has coefficients in $\{0, \pm 1\}$ and is a CW *of weight $k_1 k_2$ in G. If moreover A is proper in H, B is proper in K' and $H + K' = G$, then AB is proper in G.*

Proof. If $h + b = h' + b'$ with $h, h' \in \operatorname{supp} A$ and $b, b' \in \operatorname{supp} B$, then $h - h' = b' - b$ lies in both difference sets, so $h = h'$ and $b = b'$. Hence the $|\operatorname{supp} A| \cdot |\operatorname{supp} B|$ products are distinct, and every coefficient of AB is 0 or ± 1. Also $(AB)(AB)^{(-1)} = AA^{(-1)}BB^{(-1)} = k_1 k_2$.

Now suppose $\operatorname{supp}(AB) \subseteq c + L$ for some $L < G$. Fixing $b \in \operatorname{supp} B$ gives $\operatorname{supp} A \subseteq c - b + L$, hence $\operatorname{supp} A - \operatorname{supp} A \subseteq L \cap H$. Since A is proper in H, this forces $L \cap H = H$, i.e. $H \leq L$. Likewise $K' \leq L$, so $L \supseteq H + K' = G$. □

For coprime orders, Lemma 3.3 is the Kronecker product construction of Arasu and Seberry [2] (where $H \cap K' = 0$). The version for non-coprime orders is elementary and may well be folklore. We do not claim it as new; we only note that it was not applied to $n = 504$ in the tables.

4 The twin of the Singer-complement construction

Definition 4.1. For $c \in F$ define $W_c \colon G \to \{0, \pm 1\}$ by

$$W_c([y]) = \eta\big(\mathrm{N}(c\ \mathrm{T}(y) - y) \cdot \mathrm{T}(y)\big).$$

This is well defined: replacing y by λy ($\lambda \in F^*$) multiplies the argument of η by λ^4. As usual, W_c also denotes $\sum_g W_c(g)\, g \in \mathbb{Z}[G]$ and, via $i \mapsto [\alpha^i]$, a sequence of length m.

Lemma 4.2. $\operatorname{supp} W_0 = \operatorname{supp} W_{1/2} = S$.

Proof. Both values vanish when $\mathrm{T}(y) = 0$. Let $\mathrm{T}(y) \neq 0$. Then $\mathrm{N}(-y) \neq 0$. If $\frac{1}{2}\mathrm{T}(y) - y = 0$, then $y \in F^*$, so $\mathrm{T}(y) = 3y$ and $\frac{1}{2}\mathrm{T}(y) - y = \frac{1}{2}y \neq 0$, a contradiction. □

Proposition 4.3 (classical)**.** *W_0 is a proper* $\mathrm{CW}(m, q^2)$*.*

Proof. For $y \in P$ we have $W_0([y]) = \eta(\mathrm{N}(-y))$. Let $\delta \in K^* \setminus F^*$. Using the homogeneous formula for $W_0([\delta y])$ we get

$$\sum_{y\in P} W_0([y])W_0([\delta y]) = \sum_{y\in P} \eta(\mathrm{N}(-y))\,\eta\big(\mathrm{N}(-\delta y)\,\mathrm{T}(\delta y)\big) = \eta(\mathrm{N}\,\delta)\sum_{y\in P}\eta(\mathrm{T}(\delta y)).$$

The map $y \mapsto \mathrm{T}(\delta y)$ is affine on the affine plane P. It is not constant: otherwise $\mathrm{T}(\delta z) = 0$ for all $z \in \ker \mathrm{T}$, so $\mathrm{T}(\delta\,\cdot) = \lambda\,\mathrm{T}$ for some $\lambda \in F$, and nondegeneracy of the trace form gives $\delta = \lambda \in F$. A nonconstant affine function on $\mathbb{F}_q^2$ takes every value exactly q times, and $\sum_{t\in F}\eta(t) = 0$. So all nontrivial autocorrelations vanish, and the trivial one equals $|S| = q^2$. Finally, a coset of a proper subgroup of $\mathbb{Z}_m$ has at most $m/3 < q^2$ elements, so W_0 is proper. □

This is a standard construction of a $\mathrm{CW}(q^2 + q + 1, q^2)$; cf. [2, Theorem 4] and the RDS constructions in [3]. Our object of interest is $W_{1/2}$.

Remark 4.4 (geometry). Let $\psi(y) = \mathrm{T}(y) - 2y$. This map is F-linear and invertible, with $\psi^{-1}(u) = \frac{1}{2}(\mathrm{T}(u) - u)$; it satisfies $\mathrm{T}\circ\psi = \mathrm{T}$. On $\mathrm{PG}(2, q)$ it acts as the homology with axis L ($\psi = -2\,\mathrm{id}$ on $\ker \mathrm{T}$) and centre $[1]$ ($\psi(1) = 1$); for $p = 3$ one has $-2 = 1$ and $[1] \in L$, so ψ is then an elation. One checks that $W_{1/2} = \eta(-2)\cdot(W_0 \circ \psi)$. In conjugate coordinates (y, y^q, y^{q^2}), the cubic $\mathrm{N}(y)$ vanishes on the "Singer triangle" of three conjugate lines. The map ψ sends this triangle to its medial triangle, so $W_{1/2}$ is η of the quartic $(a+b+c)(-a+b+c)(a-b+c)(a+b-c)$ (Heron's product), while W_0 is η of $abc(a + b + c)$, up to constants.

Lemma 4.5 (Fourier form)**.** *Let χ be a character of K^* that is trivial on F^*, i.e. a character of G. Then*

$$\chi(W_0) = \eta(-1)\sum_{y\in P}\eta_K(y)\chi(y), \qquad \chi(W_{1/2}) = \eta(2)\,J_P(\chi),$$

where

$$J_P(\chi) := \sum_{w\in K,\ \mathrm{T}(w)=1} \eta_K(w)\,\chi(1-w).$$

Hence $W_{1/2}$ is a $\mathrm{CW}(m, q^2)$ *if and only if $|J_P(\chi)| = q$ for every such χ.*

Proof. Use the representatives $y \in P$, for which $\mathrm{T}(y) = 1$. Then $W_{1/2}([y]) = \eta(\mathrm{N}(\frac{1}{2} - y))$. Substitute $w = 1 - 2y$; this is a bijection of P onto itself since $\mathrm{T}(1) = 3$. Then $\frac{1}{2} - y = \frac{1}{2}w$ and $\mathrm{N}(\frac{1}{2}w) = \mathrm{N}(w)/8$, so $\eta(\mathrm{N}(\frac{1}{2}w)) = \eta(2)\eta_K(w)$. Moreover $\chi(y) = \chi(\frac{1}{2}(1 - w)) = \chi(1 - w)$, because χ is trivial on F^*. The formula for W_0 is immediate. Finally apply Lemma 2.2. □

The sum $J_P(\chi)$ is a Jacobi sum restricted to the hyperplane $\mathrm{T} = 1$. The analogous restricted Gauss sum $\sum_{y\in P}\eta_K(y)\chi(y)$ has absolute value q by Proposition 4.3. We could not evaluate $J_P(\chi)$ in closed form. Numerically (for $q = 7$), $\chi(W_{1/2})/q$ is neither a root of unity times a Galois conjugate of $\chi(W_0)$, nor a simple product or quotient of such values.

Conjecture 4.6 (twin conjecture)**.** *For every odd prime power q, $W_{1/2}$ is a* $\mathrm{CW}(q^2 + q + 1, q^2)$*. Equivalently, $|J_P(\chi)| = q$ for every character χ of K^* trivial on F^*.*

Theorem 4.7. *Let q be an odd prime power for which Conjecture 4.6 holds and $W_{1/2} \neq \pm W_0$. Define $A \in \{0, \pm 1\}^{2m}$ by*

$$A_x = \begin{cases} \frac{1}{2}(W_0(x \bmod m) + W_{1/2}(x \bmod m)), & x \text{ even}, \\ \frac{1}{2}(W_0(x \bmod m) - W_{1/2}(x \bmod m)), & x \text{ odd}. \end{cases}$$

Then A is a proper $\mathrm{CW}(2(q^2+q+1), q^2)$. The hypotheses hold, and hence such an A exists, for every odd prime power $q \leq 251$. For $q = 11, 13, 17, 19$ the resulting matrices $\mathrm{CW}(266, 11^2)$, $\mathrm{CW}(366, 13^2)$, $\mathrm{CW}(614, 17^2)$ and $\mathrm{CW}(762, 19^2)$ are certified in Lean (§7).

Proof. By Lemma 4.2, Proposition 4.3 and the hypothesis, W_0 and $W_{1/2}$ are $\mathrm{CW}(m, q^2)$s with the same support, and W_0 is proper. Now apply Corollary 3.2. The verification for $q \leq 127$ is described in Remark 4.8. □

Remark 4.8 (computational evidence).

1. For each of the 37 odd prime powers $q \leq 127$ we built $\mathbb{F}_{q^3}$ from a primitive polynomial and computed W_0, $W_{1/2}$ and A. All periodic autocorrelations were computed exactly. We used integer sequences and an FFT, checking that every value lay within 10^{-3} of an integer before rounding; for $q \leq 19$ we also used direct $O(k^2)$ integer summation. In every case W_0 and $W_{1/2}$ are $\mathrm{CW}(m, q^2)$s with the same support, $W_{1/2} \neq \pm W_0$, and A is a proper $\mathrm{CW}(2m, q^2)$. A second, independently written implementation (`data/twin_check_gf.py`, which builds $\mathbb{F}_{q^3}$ as $\mathbb{F}_p[x]/(h)$ for every prime power $q = p^k$ and uses its own primitive polynomials) confirms all of this for the 62 odd prime powers $q \leq 251$ (`data/twin_family_check_q_le_251.log`); for $q \leq 19$ its sequences are multiplier-equivalent to the certified ones, and $W_{1/2}$ is never of the form $\pm W_0(tx + s)$.

2. For $q = 3, 5, 7$, among all ± 1 sign patterns on S that are constant on orbits of the Frobenius multiplier $y \mapsto y^q$, the only ones giving a CW are $\pm W_0$ and $\pm W_{1/2}$.

3. Among the W_c, $c \in F$, only $c \in \{0, \frac{1}{2}\}$ give a CW for $q = 5, 7, 11, 13$.

4. For $q = 5$, exactly 186 of the 372000 elements $g \in \mathrm{PGL}(3, 5)$ make $W_0 \circ g$ a CW. Every $g \in N(C) \cup \psi N(C)$ does so, where $N(C)$ is the normaliser of the Singer group C ($|N(C)| = 3m = 93$). These two cosets are disjoint, because ψ fixes all $q+1$ points of L while every non-identity element of $N(C)$ fixes at most 3 points. Hence the 186 elements are exactly $N(C) \cup \psi N(C)$: up to the obvious symmetries, the twin is the only companion.

5. For the even prime power $q = 8$, no Frobenius-invariant same-support companion of the classical $\mathrm{CW}(73, 64)$ exists (the construction needs q odd).

6. A naive quartic-character analogue, intended to give $\mathrm{CW}(t(q^2+q+1), q^2)$ for $t \mid q-1$ with $t > 2$, fails for $t = 4$ ($q = 5, 9, 13$) and for $t = 5, 10$ ($q = 11$).

The certified rows for $q = 11, 13, 17, 19$ use primitive polynomials x^3+10x^2+6x+9, x^3+12x^2+9x+2, $x^3+7x^2+8x+14$ and $x^3+2x^2+18x+4$ respectively (with $\alpha = x$). The file `data/twin_singer.py` rebuilds them from Definition 4.1 in pure Python and checks equality with the certificates.

Remark 4.9. For $q = 3, 5, 7, 9$, Theorem 4.7 produces proper $\mathrm{CW}(26, 3^2)$, $\mathrm{CW}(62, 5^2)$, $\mathrm{CW}(114, 7^2)$ and $\mathrm{CW}(182, 9^2)$. In [3] these orders appear as sporadic entries found by earlier searches. We have not checked whether our matrices are equivalent to the tabulated ones. In terms of [1, Theorem 6.6],

$2(q^2+q+1) = (q^3-1)/n$ with $n = (q-1)/2$. So the twin construction re-establishes exactly this case of the withdrawn claim: unconditionally for the certified values of q, and conjecturally in general. The other withdrawn orders, for example $364 = 4 \cdot 91$ ($q = 9$) and $732 = 4 \cdot 183$ ($q = 13$), remain open.

5 A proper $\mathrm{CW}(146, 8^2)$

Here $146 = 2 \cdot 73$ with $73 = 8^2 + 8 + 1$, but $q = 8$ is even and §4 does not apply. Write a putative $\mathrm{CW}(146, 64)$ as $A = U + zV$ as in Lemma 3.1. Then $B = U + V$ and $B' = U - V$ satisfy $BB^{(-1)} = B'B'^{(-1)} = 64$ and have entries in $\{0, \pm 1, \pm 2\}$. In other words, they are integer weighing matrices $\mathrm{ICW}_2(73, 64)$, with $B \equiv B' \pmod 2$ and $|B \pm B'| \leq 2$ entrywise. Conversely, any such pair gives a $\mathrm{CW}(146, 64)$ by Lemma 3.1.

By McFarland's multiplier theorem in the form used in [1, Theorem 4.1] ($\gcd(73, 64) = 1$, $64 = 2^6$), 2 is a multiplier of B and of B'. We only used this to restrict the search: we required B to be fixed by $x \mapsto 2x$ and B' to be a translate of a 2-fixed element. The orbits of $x \mapsto 2x$ on $\mathbb{Z}_{73}$ have sizes $1, 9, 9, \ldots, 9$. Exhaustive enumeration of orbit-constant B with entries in $\{0, \pm 1, \pm 2\}$, augmentation 8 and $BB^{(-1)} = 64$ gives 32 solutions. Pairing them with all translates and signs and keeping $\{0, \pm 1\}$ lifts gives 112 $\mathrm{CW}(146, 64)$, of which 96 are proper. In the certified example, the positions where B (respectively B') equals ± 2 form a planar $(73, 9, 1)$ difference set, i.e. a line of $\mathrm{PG}(2, 8)$. Unlike in §4, the supports of U and V overlap. The first row is given in Appendix A.

We ran the same search for every repository-open entry $n = 2m$ (m odd) whose McFarland multiplier group is nontrivial: 597 entries, with at most $3 \cdot 10^5$ recursion nodes per entry. It found only $n = 146$. For 375 entries it found no solution of this restricted form; the other 221 entries (including $\mathrm{CW}(546, 16^2)$, the $q = 16$ analogue) reached the node limit. The negative outcomes are *not* nonexistence proofs.

6 A proper $\mathrm{CW}(504, 12^2)$

Let $G = \mathbb{Z}_{504}$, $H = 21\mathbb{Z}_{504} \cong \mathbb{Z}_{24}$ and $K' = 8\mathbb{Z}_{504} \cong \mathbb{Z}_{63}$. Then $H \cap K' = 168\mathbb{Z}_{504}$ has order 3 and $H + K' = G$. Take the proper $\mathrm{CW}(24, 9)$ of the repository (due to Ang, Arasu, Ma and Strassler) with $P = \{1, 5, 12, 13, 17, 18\}$ and $N = \{0, 6, 9\}$, placed on H via $i \mapsto 21i$. Take the proper $\mathrm{CW}(63, 16)$ with $P = \{0, 1, 2, 4, 8, 16, 27, 32, 45, 54\}$ and $N = \{5, 10, 17, 20, 34, 40\}$, placed on K' via $j \mapsto 8j$. The difference condition of Lemma 3.3 only needs checking inside $H \cap K'$; it holds there by direct inspection (and by machine). So the product $\sum a_i b_j X^{21i+8j}$ is a proper $\mathrm{CW}(504, 144)$. Its divisors $d \geq 144$ are 168, 252 and 504. No proper $\mathrm{CW}(d, 144)$ is known for any of them (the repository lists 168 as open and 252 as nonexistent), so this is the first $\mathrm{CW}(504, 144)$ of any kind that we know of. Iterating Lemma 3.3 (and a generalized doubling) over all explicit matrices of the repository produced no other pair (n, k) missing from [3, Table 7].

7 Formal verification

The file `lean/Main.lean` (Lean 4.34.1, core library only, no Mathlib) contains the following definitions (verbatim, except that line breaks in `MainStatement` are adjusted and the two theorems are shown without their proofs). The statement uses the index form `pafIdx`; the proofs use the fast form `paf`.

```
def dot : List Int → List Int → Int
```

```
  | x :: xs, y :: ys => x * y + dot xs ys
  | _, _ => 0
def pafIdx (a : List Int) (d : Nat) : Int :=
  ((List.range a.length).map (fun i => a.getD i 0 * a.getD ((i + d) % a.length) 0)).sum
def paf (a : List Int) (d : Nat) : Int := dot a (a.rotateLeft d)
def IsCW (n k : Nat) (a : List Int) : Prop :=
  a.length = n ∧ (∀ x ∈ a, x = -1 ∨ x = 0 ∨ x = 1) ∧
  ∀ d, d < n → pafIdx a d = if d = 0 then (k : Int) else 0
def supp (a : List Int) : List Nat := (List.range a.length).filter (fun i => a.getD i 0 != 0)
def IsProper (a : List Int) : Prop :=
  ∀ e, e ≤ a.length → 1 < e → e ∣ a.length → ∃ i ∈ supp a, ∃ j ∈ supp a, i % e ≠ j % e
def MainStatement : Prop :=
    (∃ a, IsCW 146 64 a ∧ IsProper a) ∧ (∃ a, IsCW 266 121 a ∧ IsProper a) ∧
    (∃ a, IsCW 366 169 a ∧ IsProper a) ∧ (∃ a, IsCW 504 144 a ∧ IsProper a) ∧
    (∃ a, IsCW 614 289 a ∧ IsProper a) ∧ (∃ a, IsCW 762 361 a ∧ IsProper a)
theorem paf_eq_pafIdx (a : List Int) (d : Nat) (hd : d < a.length) :
    paf a d = pafIdx a d                        -- proved in general in the file
theorem new_proper_circulant_weighing_matrices : MainStatement   -- proved at the end
```

Each witness `cwX` (an explicit list of integers) comes with `cwX_isCW : IsCW n k cwX` and `cwX_proper : IsProper cwX`. Both are proved by `decide +kernel`. For the autocorrelation, the shift range is split into blocks of 64 to bound kernel memory, the blocks are combined with `omega`, and the result is transported to `pafIdx` by the general lemma `paf_eq_pafIdx`. That lemma is proved from `dot_eq_sum` and `rotateLeft_getD`. A second theorem, `twin_singer_family_members`, certifies the eight explicit rows of Theorem 4.7 for $q = 3, 5, 7, 9, 11, 13, 17, 19$.

Statement audit.

- Since $n =$ `a.length`, all indices in `pafIdx` are $< n$, so `getD` never returns its default. Thus `pafIdx a d` is literally $\sum_{i<n} a_i a_{(i+d) \bmod n}$, and `IsCW` quantifies over all shifts $0 \le d < n$, with value k at $d = 0$ and 0 otherwise. For the circulant matrix $W_{ij} = a_{(j-i) \bmod n}$ one has $(WW^{\mathsf{T}})_{ij} = \mathrm{PAF}_a((i-j) \bmod n)$, so this is exactly $WW^{\mathsf{T}} = kI$. (This last standard translation is not formalised.)

- `IsProper`: the subgroups of $\mathbb{Z}_n$ are $e\mathbb{Z}_n$ with $e \mid n$, proper exactly when $e > 1$, and a set lies in a coset of $e\mathbb{Z}_n$ exactly when all its elements are congruent mod e. The bound $e \le n$ is automatic for divisors.

- The file contains sanity checks: $\mathrm{CW}(7, 4) = -1 + X + X^2 + X^4$ satisfies `IsCW 7 4` and `IsProper`; its lift to $\mathbb{Z}_{14}$ satisfies `IsCW 14 4` but *not* `IsProper`; and a corrupted sequence fails `IsCW`. So neither predicate is vacuous.

- The main statement has no hypotheses; it is a conjunction of closed existential claims with explicit witnesses in the file.

Axioms and resources. `lean/lean_output.txt` reports, for the main theorem and for every certificate theorem, dependence only on `propext`, `Classical.choice` and `Quot.sound` (the `_proper` theorems use only `propext`). There is no `sorryAx` and no `native_decide`. The full check took 491 s of wall time with a maximum resident memory of 2.1 GB. The certificates were also checked independently by a plain Python script (`data/verify_certificates.py`).

8 What is not verified; open questions

- **Conjecture 4.6 is open.** It is verified for odd prime powers $q \le 251$ only, and the infinite family $\mathrm{CW}(2(q^2+q+1), q^2)$ is conditional on it. A proof would amount to evaluating, or at least bounding exactly, the restricted Jacobi sums $J_P(\chi)$. Several approaches failed for us: a direct Gauss-sum expansion leads to twisted character sums of cubic polynomials; we found no collineation symmetry beyond Remark 4.8(4); and the quartic analogue fails.
- **Novelty** is judged against [4, 3, 1, 10] and a literature search on circulant weighing matrices and perfect ternary sequences. We could not consult Krengel [6]. The repository data file contains some stale entries (for example, some orders covered by infinite families are still marked open), which is why we also compared with the curated Table 7 of [3].
- The finite-field constructions and the searches are not formalised. Lean certifies only the final explicit sequences, which is all the main theorem needs.
- Open: the even case ($q = 16$ gives $\mathrm{CW}(546, 16^2)$); the other withdrawn cases of [1, Theorem 6.6] (364 for $q = 9$, 732 for $q = 13$); and the remaining open entries of Strassler's table [1, 10], none of which is touched here.

A First rows of the six new matrices

For each matrix we list $P = \{i : a_i = 1\}$ and $N = \{i : a_i = -1\}$ (0-based positions in the first row of the circulant; all other entries are 0). The same data, together with the four known-order family members, is in `data/certificates.json` and `data/certificates_PN.txt`.

$\mathrm{CW}(146, 8^2)$ **(Section 5).** Lean name `cw146`; $|P| = 36$, $|N| = 28$.
P = {5, 7, 11, 15, 21, 22, 30, 34, 39, 42, 44, 47, 60, 66, 68, 83, 84, 87, 88, 90, 93, 94, 95, 101, 103, 106, 113, 115, 117, 118, 120, 126, 129, 132, 133, 136}.
N = {0, 2, 3, 4, 8, 16, 17, 19, 23, 32, 33, 45, 53, 59, 63, 64, 74, 79, 85, 97, 107, 110, 111, 119, 121, 128, 139, 141}.

$\mathrm{CW}(266, 11^2)$ **(Theorem 4.7, $q = 11$).** Lean name `cwFam266`; $|P| = 66$, $|N| = 55$.
P = {4, 7, 8, 12, 14, 18, 24, 27, 28, 31, 32, 35, 38, 39, 40, 42, 44, 49, 50, 52, 58, 59, 75, 76, 77, 83, 86, 88, 91, 98, 102, 105, 106, 107, 113, 115, 117, 119, 120, 122, 132, 134, 144, 148, 152, 154, 156, 163, 170, 174, 179, 196, 197, 198, 201, 203, 213, 215, 218, 223, 237, 244, 245, 254, 256, 264}.
N = {0, 6, 9, 13, 16, 25, 34, 36, 45, 48, 57, 66, 69, 71, 72, 74, 79, 81, 93, 95, 99, 100, 103, 108, 118, 124, 125, 130, 138, 143, 162, 176, 180, 184, 186, 188, 189, 193, 194, 200, 206, 211, 217, 222, 225, 227, 229, 234, 243, 247, 249, 259, 260, 261, 262}.

$\mathrm{CW}(366, 13^2)$ **(Theorem 4.7, $q = 13$).** Lean name `cwFam366`; $|P| = 91$, $|N| = 78$.
P = {2, 3, 12, 18, 20, 23, 25, 26, 31, 34, 35, 37, 39, 46, 48, 59, 60, 69, 76, 86, 88, 89, 95, 101, 105, 106, 107, 109, 112, 114, 115, 121, 129, 131, 137, 138, 141, 149, 156, 160, 165, 177, 179, 183, 190, 194, 197, 198, 199, 207, 212, 213, 215, 216, 227, 232, 233, 234, 239, 246, 249, 250, 256, 258, 260, 262, 264, 267, 268, 270, 274, 275, 280, 281, 293, 299, 309, 315, 317, 319, 322, 325, 326, 330, 338, 346, 353, 357, 358, 359, 365}.
N = {4, 5, 6, 9, 19, 21, 22, 36, 41, 42, 52, 53, 54, 57, 58, 65, 68, 70, 71, 78, 80, 93, 94, 96, 102, 111, 113, 117, 118, 120, 123, 124, 133, 135, 144, 146, 148, 150, 151, 152, 157, 164, 166, 167, 173, 178, 180, 181, 184, 191, 196, 211, 223, 228, 238, 247, 255, 265, 266, 273, 282, 283, 286, 287, 291, 302, 308, 310, 323, 328, 336, 337, 341, 342, 344, 345, 352, 355}.

$\mathrm{CW}(504, 12^2)$ **(Section 6).** Lean name `cw504`; $|P| = 78$, $|N| = 66$.
P = {2, 4, 5, 21, 25, 29, 33, 37, 40, 53, 69, 80, 85, 90, 105, 108, 109, 113, 121, 129, 130, 136, 137, 149, 160, 166, 169, 180, 201, 206, 213, 229, 233, 234, 237, 252, 260, 262, 268, 269, 272, 273, 277, 281, 284, 285, 286, 289, 305, 306, 316,

320, 321, 325, 337, 349, 357, 361, 365, 373, 378, 380, 381, 386, 389, 394, 398, 401, 410, 421, 442, 446, 453, 461, 465, 468, 485, 489}.
N = {0, 8, 10, 13, 16, 20, 32, 34, 41, 45, 54, 61, 64, 68, 89, 101, 117, 125, 126, 128, 134, 142, 145, 146, 157, 158, 173, 181, 185, 189, 190, 194, 197, 205, 216, 221, 241, 253, 254, 256, 265, 292, 293, 313, 317, 332, 341, 342, 353, 360, 377, 382, 388, 397, 405, 409, 412, 418, 425, 432, 433, 437, 445, 458, 486, 493}.

CW$(614, 17^2)$ **(Theorem 4.7,** $q = 17$**).** Lean name `cwFam614`; $|P| = 153$, $|N| = 136$.
P = {2, 5, 6, 9, 10, 12, 13, 18, 27, 29, 34, 50, 55, 56, 60, 62, 73, 78, 84, 85, 88, 93, 98, 102, 104, 105, 107, 112, 118, 121, 122, 124, 129, 135, 141, 143, 145, 148, 153, 158, 159, 160, 163, 164, 167, 170, 173, 175, 176, 180, 189, 190, 200, 204, 207, 215, 217, 220, 221, 223, 224, 225, 226, 227, 230, 232, 236, 237, 240, 242, 247, 252, 258, 259, 260, 263, 264, 265, 266, 268, 273, 274, 286, 288, 290, 305, 306, 308, 315, 321, 324, 328, 330, 332, 333, 338, 339, 342, 343, 345, 351, 353, 359, 360, 371, 376, 378, 383, 396, 397, 398, 399, 406, 430, 434, 435, 438, 440, 441, 443, 444, 449, 453, 459, 475, 485, 493, 506, 515, 516, 519, 529, 536, 540, 545, 555, 556, 557, 564, 577, 578, 584, 585, 589, 591, 592, 594, 596, 598, 599, 600, 604, 609}.
N = {0, 4, 7, 19, 20, 22, 39, 41, 43, 47, 49, 57, 58, 61, 68, 75, 77, 79, 80, 81, 83, 94, 97, 101, 103, 108, 113, 115, 117, 119, 120, 125, 126, 132, 138, 140, 144, 147, 149, 150, 155, 156, 157, 169, 172, 179, 181, 183, 184, 185, 188, 195, 196, 198, 203, 210, 213, 214, 218, 219, 241, 245, 246, 254, 262, 267, 269, 275, 276, 283, 295, 296, 300, 310, 322, 323, 331, 335, 337, 340, 344, 352, 355, 358, 361, 370, 372, 374, 377, 381, 393, 394, 402, 403, 413, 417, 418, 421, 423, 437, 446, 458, 468, 478, 481, 484, 489, 498, 499, 500, 501, 504, 509, 512, 513, 518, 523, 535, 538, 541, 542, 550, 551, 558, 560, 562, 563, 568, 579, 586, 587, 588, 601, 606, 608, 611}.

CW$(762, 19^2)$ **(Theorem 4.7,** $q = 19$**).** Lean name `cwFam762`; $|P| = 190$, $|N| = 171$.
P = {0, 6, 7, 9, 11, 14, 29, 35, 43, 47, 53, 55, 56, 60, 66, 70, 78, 80, 83, 92, 94, 96, 97, 98, 99, 100, 103, 105, 114, 115, 124, 130, 131, 132, 133, 135, 138, 140, 142, 145, 147, 151, 153, 158, 161, 171, 174, 178, 184, 191, 193, 195, 197, 199, 201, 203, 204, 208, 209, 211, 212, 218, 219, 221, 222, 224, 227, 229, 241, 245, 248, 250, 251, 257, 258, 262, 266, 267, 271, 278, 279, 283, 286, 288, 291, 295, 300, 302, 303, 311, 319, 320, 324, 326, 330, 331, 332, 334, 336, 338, 344, 351, 357, 358, 360, 366, 367, 369, 371, 373, 374, 375, 376, 378, 382, 389, 396, 398, 400, 401, 404, 406, 408, 409, 412, 413, 417, 420, 422, 423, 433, 440, 443, 446, 448, 458, 460, 462, 469, 471, 482, 489, 492, 501, 503, 507, 529, 533, 536, 541, 551, 563, 567, 568, 573, 575, 577, 581, 583, 597, 607, 619, 621, 633, 642, 650, 657, 661, 665, 666, 675, 686, 687, 695, 704, 706, 710, 716, 720, 726, 727, 729, 733, 740, 742, 743, 744, 746, 758, 761}.
N = {3, 4, 12, 16, 18, 22, 24, 26, 45, 49, 50, 51, 54, 57, 58, 68, 71, 73, 76, 87, 93, 107, 109, 117, 119, 123, 129, 134, 136, 137, 139, 144, 154, 159, 163, 164, 165, 168, 169, 173, 175, 179, 181, 183, 185, 188, 198, 207, 213, 214, 228, 232, 233, 235, 237, 239, 242, 243, 249, 253, 255, 256, 260, 263, 264, 265, 270, 272, 273, 277, 282, 287, 292, 293, 297, 298, 304, 307, 309, 316, 317, 321, 322, 327, 328, 337, 340, 341, 342, 343, 353, 355, 364, 368, 379, 383, 386, 391, 402, 415, 418, 419, 421, 425, 429, 442, 444, 445, 450, 455, 456, 463, 466, 467, 470, 472, 476, 485, 487, 491, 494, 499, 506, 509, 522, 524, 527, 530, 531, 537, 538, 547, 548, 553, 557, 558, 561, 571, 586, 587, 591, 596, 598, 601, 611, 612, 615, 617, 625, 640, 649, 655, 656, 662, 670, 671, 677, 682, 689, 691, 693, 694, 696, 699, 714, 730, 731, 735, 737, 751, 753}.

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